

Functions

From "formula" to "function": a rule is not enough — you must say where it starts and where it lands.

Session	Date	Today
3 of 15	Wed 9 Sep 2026	Domain, codomain, undoing

Where session 3 sits

1 · Course overview and school-maths refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Elementary combinatorics and binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY Domain and codomain, injective/surjective/bijective, composition and inverses, and the unit circle.

Set theory, in one page

OBJECTS		OPERATIONS	
set	Unordered, no repeats: $\{1, 2, 2\} = \{1, 2\}$	$A \cup B$	Union: A or B (inclusive)
$x \in A$	Element of a set	$A \cap B$	Intersection: in both. Empty: disjoint
$\{1, 2, 3\}$	Roster ; or set-builder $\{x \in \mathbb{Z} : x^2 < 10\}$	$A \setminus B$	Difference: A without B . Asymmetric
$\emptyset$	Empty: no elements. $\{\emptyset\}$ has one	A^c	Complement $U \setminus A$; name U first
$A \subseteq B$	Subset: all of A lies in B	$A \triangle B$	Symmetric difference: exactly one
$A = B$	$A \subseteq B$ and $B \subseteq A$: double inclusion	De Morgan	$(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$
$\mathcal{P}(A)$	Power set: all subsets. $ \mathcal{P}(A) = 2^{ A }$	$A \times B$	Ordered pairs (a, b) ; $ A \times B = A B $

TODAY'S QUESTION Is x^2 a function? Can you run it backwards?

A function is three things, not one

DEFINITION A function $f : A \rightarrow B$ is three things at once:

- a **domain** A
- a **codomain** B
- a rule giving **each** $a \in A$ **exactly one** $f(a) \in B$

Its **graph** is a subset of $A \times B$, with each a occurring in exactly one pair.

IS $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$. Every real has exactly one square, and it is real.

IS NOT

$g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = 1/x$: nothing at $x = 0$. *Not total*.

The relation $y^2 = x$ on $\mathbb{R}$: two possible y values when $x = 4$, and none when $x = -1$. *Not a function*.

The codomain is declared; the image is computed

DEFINITION

$$f(A) = \{ f(a) : a \in A \} \subseteq B$$

$$f^{-1}(S) = \{ a \in A : f(a) \in S \}$$

The first is the **image**, the second the **preimage** of $S \subseteq B$.

WORKED

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2.$$

Codomain $\mathbb{R}$, image $[0, \infty)$. So -1 sits in the codomain but never in the image.

$$f^{-1}(\{4\}) = \{-2, 2\}, \text{ while } f^{-1}([-1, 0)) = \emptyset.$$

NOTATION WARNING

$f^{-1}(S)$ takes a **set** and returns a **set**. It exists for every function. It is not the inverse function f^{-1} we meet after the break, which needs a bijection.

Which of the three ingredients is missing?

ACTIVITY 1 GROUPS OF THREE

12 MIN

For each line: is it a function? If not, name **exactly** which requirement fails, and give the input that breaks it.

1. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n/2$
2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$
3. $f : \mathbb{Q} \rightarrow \mathbb{Z}, f(p/q) = p$
4. $f : \mathbb{R} \rightarrow [0, \infty), f(x) = x^2$
5. $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by the condition $f(x)^2 = x$

Name the requirement that fails

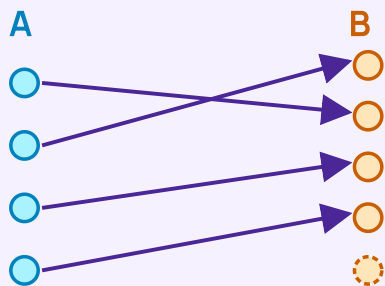
RULE	VERDICT	THE BREAKING INPUT
$1 \cdot \mathbb{Z} \rightarrow \mathbb{Z}, n/2$	Leaves the codomain	$f(3) = 1.5 \notin \mathbb{Z}$.
$2 \cdot \mathbb{R} \rightarrow \mathbb{R}, \sqrt{x}$	Not total	No value at $x = -1$.
$3 \cdot \mathbb{Q} \rightarrow \mathbb{Z}, p/q \mapsto p$	Not well defined	$\frac{1}{2} = \frac{2}{4}$, yet the rule returns 1 and 2.
$4 \cdot \mathbb{R} \rightarrow [0, \infty), x^2$	A function	Total, single-valued, lands in $[0, \infty)$.
$5 \cdot \mathbb{R} \rightarrow \mathbb{R}, f(x)^2 = x$	No such function	At $x = -1$ we would need $f(-1)^2 = -1$.

LINE 3 IS THE NEW IDEA The input is a *rational number*, not a pair of integers. A rule using p and q must agree on every fraction naming that number.

Look at the arrows arriving in B

INJECTIVE

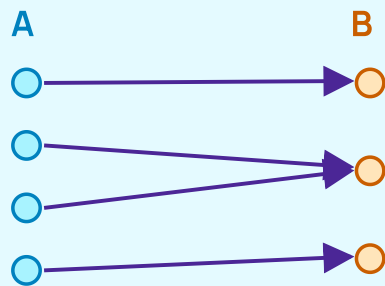
No output is hit twice.



One B is unused, so this example is **not** surjective.

SURJECTIVE

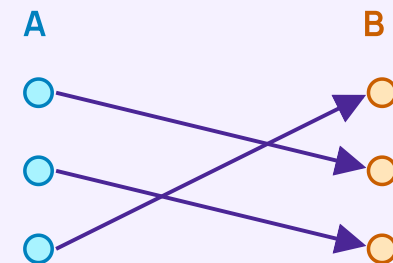
No output is missed.



The middle B is hit twice, so this example is **not** injective.

BIJECTIVE

Every output is hit exactly once.



It is injective **and** surjective.

ONE RULE TO REMEMBER

Look only at the dots in B : injective means **at most one** arrow arrives; surjective means **at least one** arrives; bijective means **exactly one** arrives.

Each of the four claims has one fixed opening move

THE FOUR TEMPLATES

Injective: assume $f(a_1) = f(a_2)$, derive $a_1 = a_2$.

Not injective: exhibit one pair $a_1 \neq a_2$ with equal images.

Surjective: take arbitrary $b \in B$, *construct* an a and check $f(a) = b$.

Not surjective: exhibit one b and *prove* nothing maps to it. ■

WORKED: $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 5$

Injective: $3a_1 - 5 = 3a_2 - 5 \Rightarrow 3a_1 = 3a_2 \Rightarrow a_1 = a_2$.

Surjective: given y , set $x = \frac{y+5}{3}$; then $f(x) = 3 \cdot \frac{y+5}{3} - 5 = y$. So f is a bijection.

Classify, then prove it

ACTIVITY 2 GROUPS OF THREE

15 MIN

For each: classify as injective, surjective, both or neither — then prove it with a template.

1. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 2n + 1$
2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$
3. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
4. $f : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n + 1$
5. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = a - b$
6. $f : \mathbb{R} \rightarrow (0, \infty), f(x) = 2^x$

Six classifications, each with its reason

f	INJECTIVE?	SURJECTIVE?	THE REASON
$1 \cdot \mathbb{Z} \rightarrow \mathbb{Z}, 2n + 1$	Yes	No	Strictly increasing; but $2n + 1$ is odd, so 2 is missed.
$2 \cdot \mathbb{R} \rightarrow \mathbb{R}, x^3$	Yes	Yes	Bijective , inverse $\sqrt[3]{y}$.
$3 \cdot \mathbb{R} \rightarrow \mathbb{R}, x^2$	No	No	$f(-1) = f(1)$; and -1 is never attained.
$4 \cdot \mathbb{N} \rightarrow \mathbb{N}, n + 1$	Yes	No	Nothing maps to 1 (taking $\mathbb{N}$ to start at 1).
$5 \cdot \mathbb{Z}^2 \rightarrow \mathbb{Z}, a - b$	No	Yes	$f(1, 1) = f(2, 2)$; every k comes from $(k, 0)$.
$6 \cdot \mathbb{R} \rightarrow (0, \infty), 2^x$	Yes	Yes	Bijective , inverse $\log_2 y$.

WATCH OUT

"The function x^2 " names no function at all

DOMAIN	CODOMAIN	INJECTIVE?	SURJECTIVE?	SO IT IS...
$\mathbb{R}$	$\mathbb{R}$	No (± 2)	No (-1)	neither
$[0, \infty)$	$\mathbb{R}$	Yes	No (-1)	injective only
$\mathbb{R}$	$[0, \infty)$	No (± 2)	Yes	surjective only
$[0, \infty)$	$[0, \infty)$	Yes	Yes	bijjective

TRAP One rule, four functions, four different answers. A question asking "is x^2 injective?" is not yet a question.

FIX Write $f : A \rightarrow B$ **before** you write the rule. Every answer you write this term starts that way.

PART II

Composing, and undoing

Functions are things you can chain. Some chains can be reversed — and the condition for that is exactly the vocabulary from the first hour.



Composition needs the codomain of one to be the domain of the next

DEFINITION The **composition** of $f : A \rightarrow B$ with $g : B \rightarrow C$ is

$$(g \circ f)(a) = g(f(a)), \quad g \circ f : A \rightarrow C$$

Read it right to left: f runs first. If the middle sets do not match, the composition does not exist.

ORDER IS NOT A DETAIL

$f(x) = x + 1, g(x) = x^2$, both $\mathbb{R} \rightarrow \mathbb{R}$.

$$(g \circ f)(x) = (x + 1)^2 \quad (f \circ g)(x) = x^2 + 1$$

At $x = 1$: 4 against 2. So $g \circ f \neq f \circ g$.

Injectivity of a chain is inherited by its first link

THEOREM Let $f : A \rightarrow B$ and $g : B \rightarrow C$. If $g \circ f$ is injective, then f is injective.

PROOF Suppose $f(a_1) = f(a_2)$. Applying g to both sides gives $g(f(a_1)) = g(f(a_2))$, that is $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g \circ f$ is injective, $a_1 = a_2$. Hence f is injective. ■

AND g ? Nothing follows. With $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, $f(1) = 1$, $g(1) = g(2) = 1$: the chain is injective, g is not.

A function can be undone exactly when it is a bijection

THEOREM $f : A \rightarrow B$ has a $g : B \rightarrow A$ with $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$ **if and only if** f is a bijection. This g is unique: it is f^{-1} .

PROOF ($\Leftarrow$) Let $b \in B$. Surjectivity gives an a with $f(a) = b$; injectivity says it is the only one. So $g(b) = a$ is well defined, and both composites are identities. ■

TRAP f^{-1} is not $\frac{1}{f}$. For $f(x) = 3x - 5$: $f^{-1}(1) = 2$, while $\frac{1}{f(1)} = -\frac{1}{2}$.

FIX f^{-1} undoes; $\frac{1}{f}$ divides. Same superscript, unrelated operations.

Find the broken step

ACTIVITY 3 GROUPS OF THREE

14 MIN

Claim. If $g \circ f$ is injective, then g is injective. **"Proof."**

- (i) Suppose $g(b_1) = g(b_2)$.
 - (ii) Write $b_1 = f(a_1)$ and $b_2 = f(a_2)$.
 - (iii) Then $(g \circ f)(a_1) = (g \circ f)(a_2)$, so $a_1 = a_2$.
 - (iv) Hence $b_1 = f(a_1) = f(a_2) = b_2$. ■
1. Which of (i)–(iv) is the first unjustified line?
 2. Give a counterexample showing the claim is false.
 3. Add the smallest hypothesis on f that repairs the proof.
 4. State and prove the companion result for $g \circ f$ *surjective*.

Step (ii) assumed what was never given

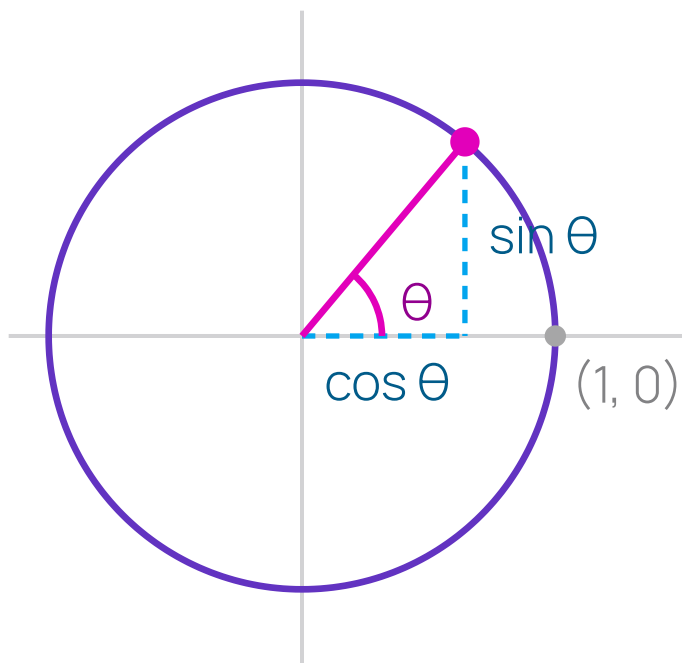
1 · THE FIRST UNJUSTIFIED LINE IS (II) "Write $b_1 = f(a_1)$ " assumes every element of B is an output of f — that is surjectivity, never assumed.

2 · COUNTEREXAMPLE $f : \{1\} \rightarrow \{1, 2\}$, $f(1) = 1$; and $g : \{1, 2\} \rightarrow \{1\}$, $g(1) = g(2) = 1$. Then $g \circ f$ is injective but g is not.

3 · THE REPAIR Assume f is **surjective**. Step (ii) is then legitimate and the rest stands.

4 · THE COMPANION RESULT If $g \circ f$ is surjective, so is g : given $c \in C$, pick a with $g(f(a)) = c$; then $f(a)$ is a preimage. ■

Sine and cosine are coordinates, and the identity is Pythagoras



DEFINITION From $(1, 0)$, walk θ counter-clockwise along the unit circle $x^2 + y^2 = 1$. The point you reach is $(\cos \theta, \sin \theta)$, and $\tan \theta = \frac{\sin \theta}{\cos \theta}$ where $\cos \theta \neq 0$. That distance is θ radians.

WHY THE IDENTITY IS FREE The point lies *on* the circle, so $x^2 + y^2 = 1$. Substitute:

$$\cos^2 \theta + \sin^2 \theta = 1$$

Not a new fact — the circle's equation, renamed. ■

Periodicity gives repeated outputs

f	DOMAIN	IMAGE	PERIOD	INJECTIVE THERE?
$\sin$	$\mathbb{R}$	$[-1, 1]$	2π	No ($\sin 0 = \sin \pi$)
$\cos$	$\mathbb{R}$	$[-1, 1]$	2π	No ($\cos(-\theta) = \cos \theta$)
$\tan$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$	$\mathbb{R}$	π	No (period π)

DEFINITION $p > 0$ is a **period** of f if $f(\theta + p) = f(\theta)$ for every θ . When there is a smallest positive one, it is the **fundamental period**.

THE RESTRICTION TRICK, AGAIN $\sin$ cut down to $[-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$ is a **bijection**, so by the theorem it has an inverse. That inverse is **arcsin**.

The unit circle definition already fixed your units

TRAP

$$\sin(30) \neq \frac{1}{2}.$$

θ is an arc length, so $\sin(30) \approx -0.988$: the point after walking 30 units round a circle of circumference 2π . What you meant was $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.

FIX

Convert once, at the start: $\theta_{\text{rad}} = \frac{\pi}{180} \theta_{\text{deg}}$.

Radians are the default here, in every calculus course, and in every programming language's standard library. Degrees are the special case that needs announcing.

THIRTY SECONDS

Convert 45° , 60° and 270° to radians. Then: what is $\cos(180)$, and what did whoever wrote it mean?

Derive the circle without us

ACTIVITY 4 PAIRS

12 MIN

Draw one unit circle. Close every other source — no tables, no phones.

1. Place $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$ and read off $(\cos \theta, \sin \theta)$ from the 30–60–90 and 45–45–90 triangles.
2. From the picture alone, argue $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$.
3. From the picture alone, argue $\sin(\theta + \pi) = -\sin \theta$.
4. 2π is a period of $\tan$. Is it the *smallest* one? Justify.

Everything read off one picture

1 · THE THREE ANGLES

$$\frac{\pi}{6} : \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) \cdot \frac{\pi}{4} : \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right) \cdot \frac{\pi}{3} : \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$$

2 · EVEN AND ODD

Negating θ **reflects the point in the x -axis**: the first coordinate is unchanged and the second flips. So $\cos(-\theta) = \cos \theta$, $\sin(-\theta) = -\sin \theta$. ■

3 · ADDING π

Adding π sends the point to its **antipode**, negating both coordinates. Reading off the second gives $\sin(\theta + \pi) = -\sin \theta$. ■

4 · "A PERIOD" IS NOT "THE PERIOD"

No. $\tan(\theta + \pi) = \tan \theta$, so π is smaller — and π is in fact the smallest. Verifying one value never establishes minimality.

On your own, before you leave

1 Choose a domain and codomain that make $x \mapsto x^2 - 4x$ a bijection, and prove both halves.

2 With $f(x) = 2x + 1$ and $g(x) = x^2$ on $\mathbb{R}$: compute $g \circ f$ and $f \circ g$, and give one x where they differ.

3 Derive $\cos \frac{\pi}{3}$ and $\sin \frac{\pi}{3}$ from the unit circle. Recalling them scores zero.

RULE FOR TODAY Every verdict carries its template. "Injective" alone is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is.

What today was actually about

- A rule is not a function. Domain and codomain are part of the object — change them and every answer changes.
- Injective and surjective are **claims to be proved**, each with a fixed opening move.
- A function is invertible exactly when it is a bijection — which is why **arcsin** needs a restricted domain.

domain

codomain

image

preimage

bijjective

composition

inverse

radian

period

Problem Set 3

Eight injective/surjective classifications, each with a proof or a counterexample, plus five unit-circle values *derived*, not recalled.

Next session: **Function families** — polynomial, exp/log and trig, and how fast each one grows.